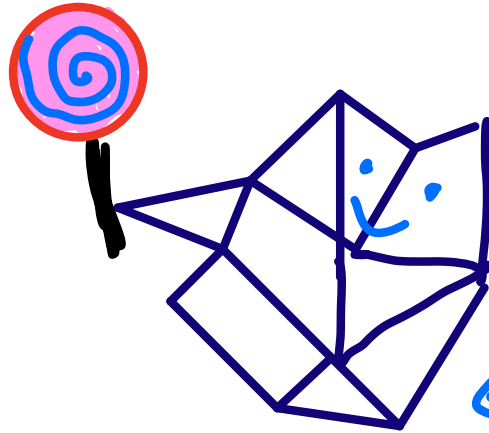


Cosmic TENSION

& Planck Scale. Messages



↑ Angry
mis-understood
Cosmos



a model of
quantum geometry
Showing Care

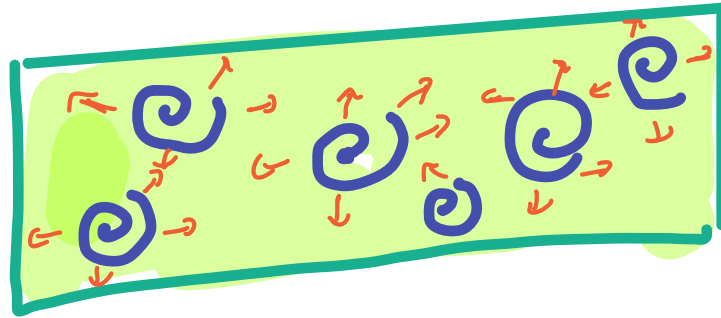
John Joseph Marchetta

"Alleviating the Tension in the Cosmic Microwave Background using Planck-Scale Physics" - Ashtekar 2020

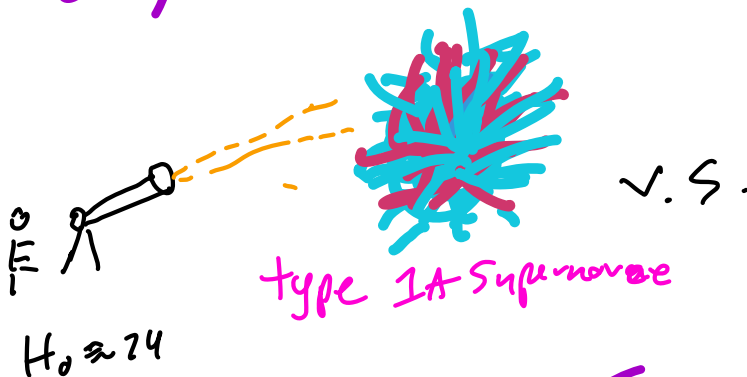
" T_0 H_0 or not to H_0 " - George Efsthion 2021

Outline

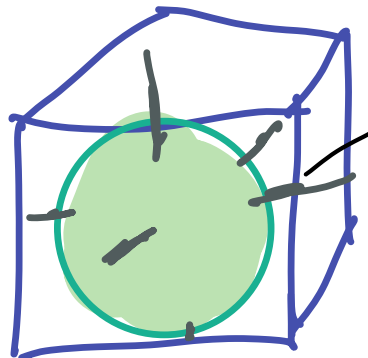
1.) Expanding Space



2.) Hubble tension

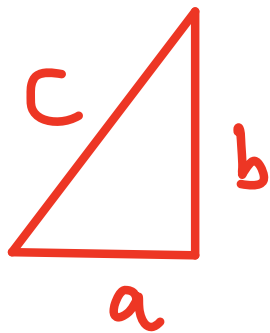


3.) Planck Era Predictions



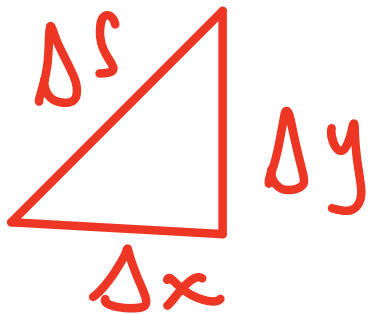
$$6 l_p^2 = \text{Area}$$

What's a Metric?



$$a^2 + b^2 = c^2$$

Physics Notation:

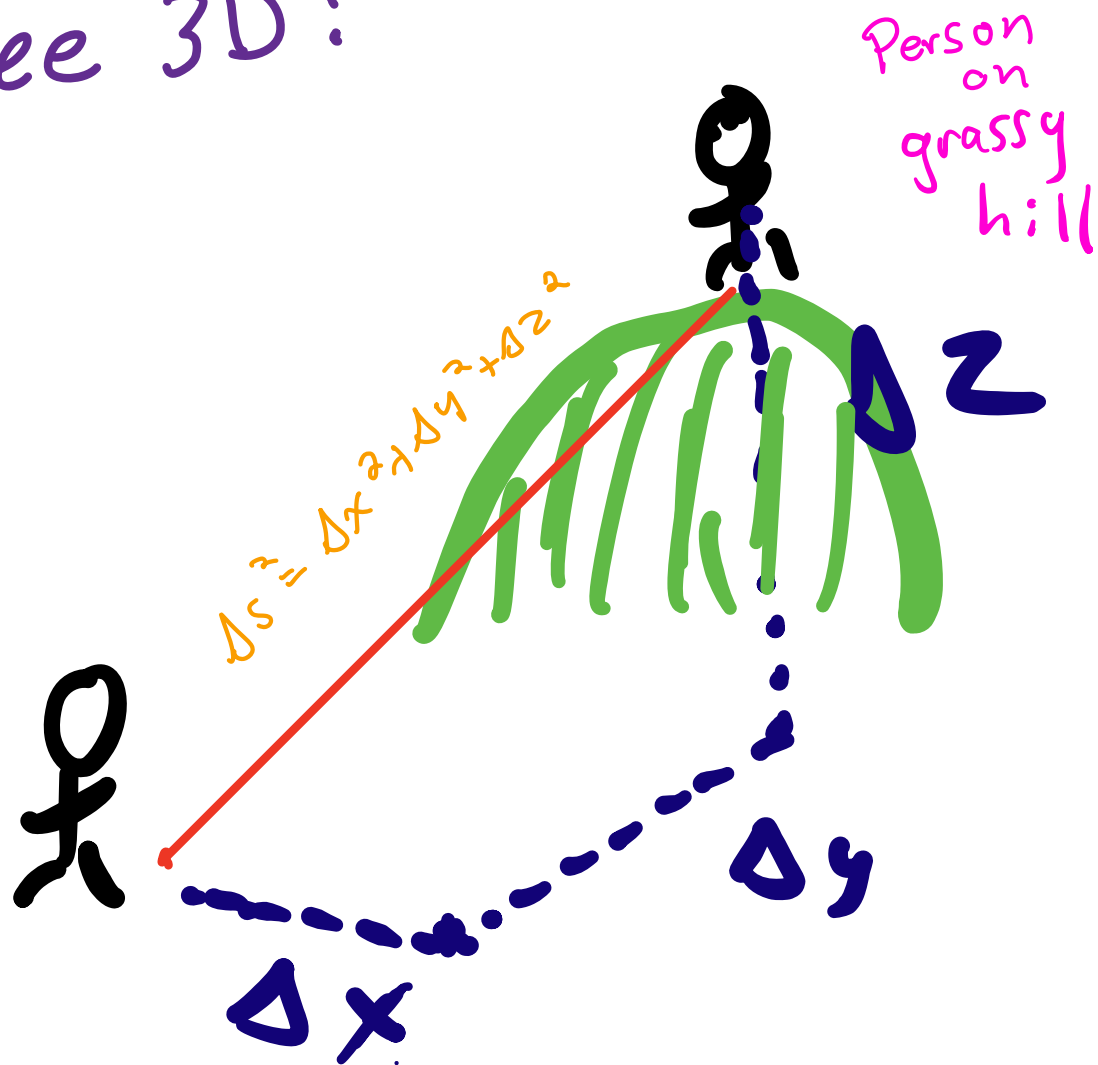


$$(\Delta x)^2 + (\Delta y)^2 = (\Delta s)^2$$

Infinitely small triangles

$$dx^2 + dy^2 = ds^2$$

Looking around we
see 3D:

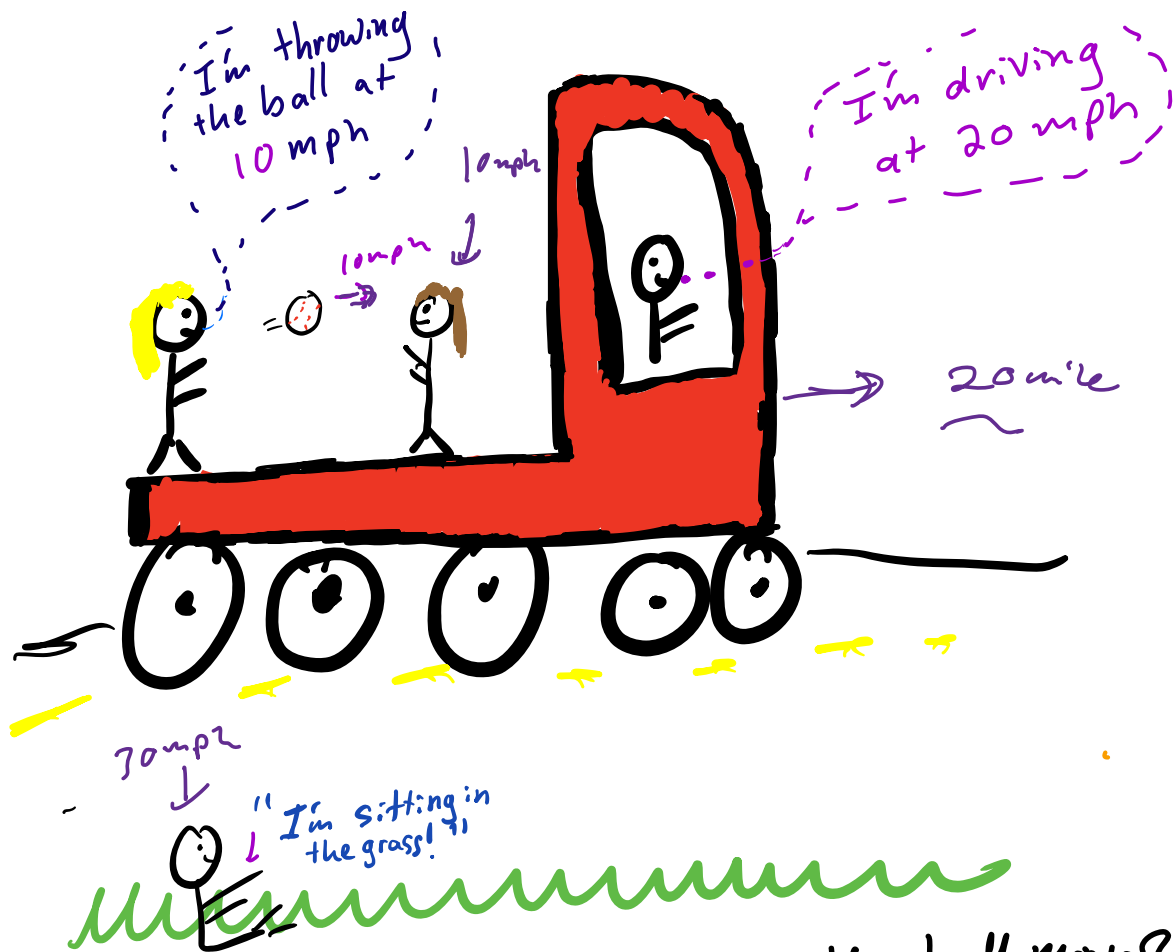


$$(\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 = (ds)^2$$

$$ds^2 = dx^2 + dy^2 + dz^2$$

Maxwell's theory: Speed of light is constant

Example of 19th Century relativity



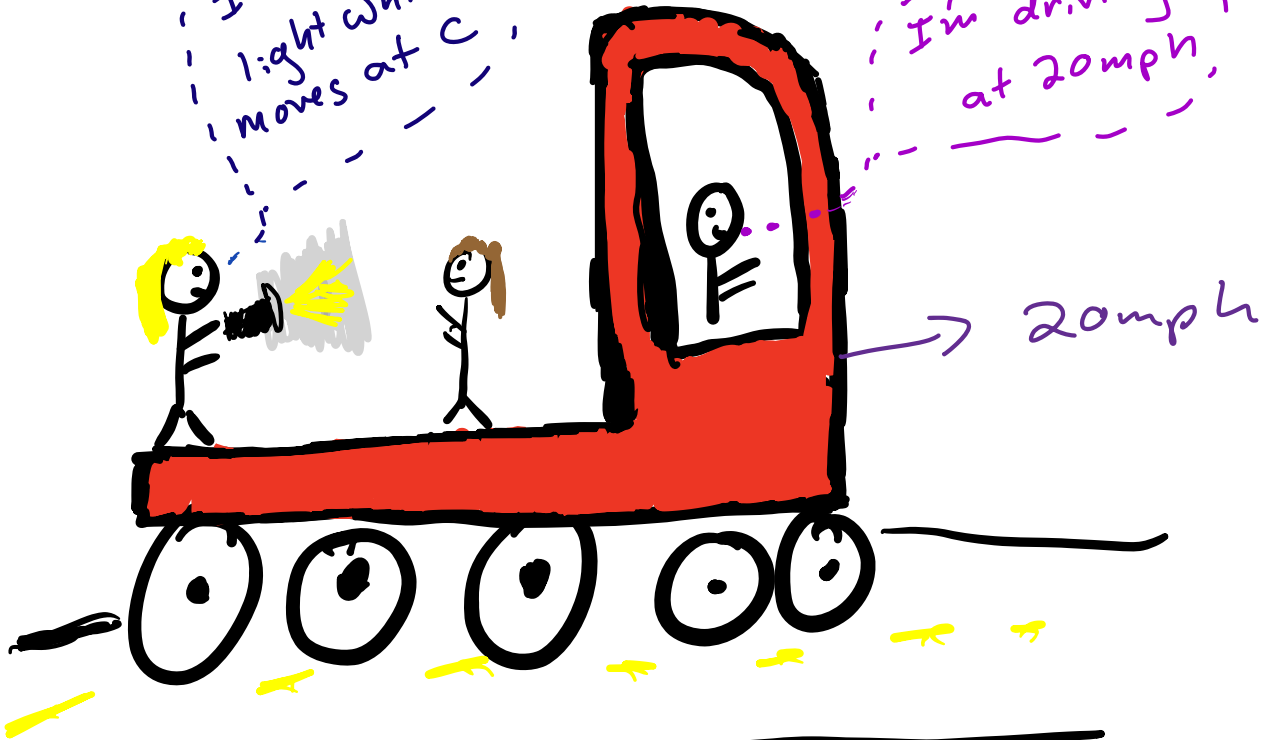
Q: what speed does the man see the ball move?

$$10 \text{ mph} + 20 \text{ mph} = 30 \text{ mph}$$

$c = 6.07 \times 10^8 \text{ mph}$

I'm shining
light which
moves at c

I'm driving
at 20mph



I'm sitting in
the grass!

$c + 20 \text{ mph} = c ?$

Relativity & Maxwell's E&M

Do Not mix

Einstein reconciled this with a
4D universe

3D: $ds^2 = dx^2 + dy^2 + dz^2$

4D: $ds^2 = dx^2 + dy^2 + dz^2 - c^2 dt^2$

~10 years later, Einstein proposed gravity was a feature of curved spacetime.

$$dx^2 + dy^2 + dz^2 \rightarrow dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

Approximating the earth to be a sphere

$$ds^2 = -\left(1 - 2\frac{GM}{c^2 r}\right) c^2 dt^2 + \frac{1}{\left(1 - 2\frac{GM}{c^2 r}\right)} dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

Newton potential



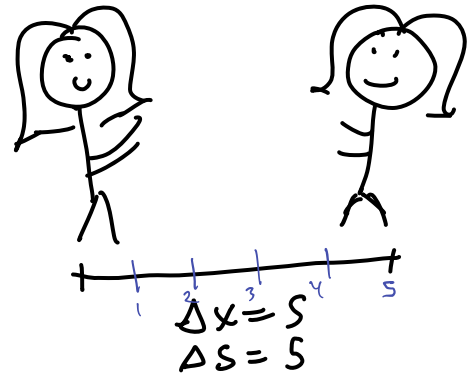
Expanding Universe Metric (FLRW)

$$(\Delta s)^2 = a(t) \left((\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 \right) - c^2 \Delta t^2$$

$$a(t) = t$$

Let two friends be
separated by $\Delta x = 5$ at

$$t = 1 \quad a(1) = 1$$



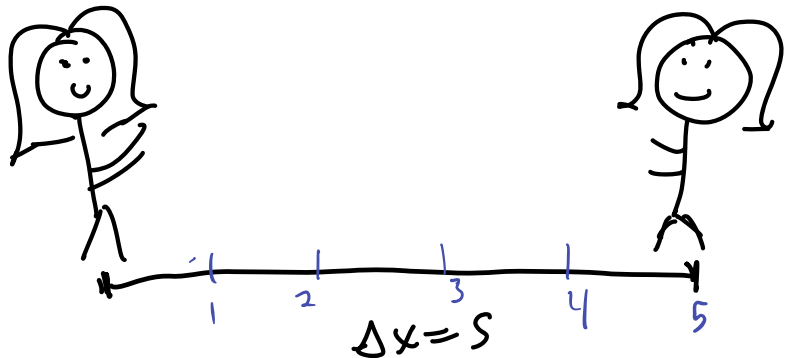
$$(\Delta s)^2 = 1 \cdot (\Delta x)^2 = 5^2 \Rightarrow \Delta s = 5$$

$$a + t = 2 \quad a(t) = t \quad q(t=2) = 2$$

$$(\Delta s)^2 = 2((\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2) - (c \Delta t)^2$$

$$(\Delta s)^2 = 2(\Delta x)^2$$

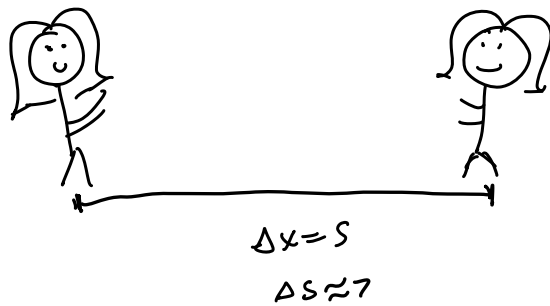
$$= 2(2s)^2 = 50$$



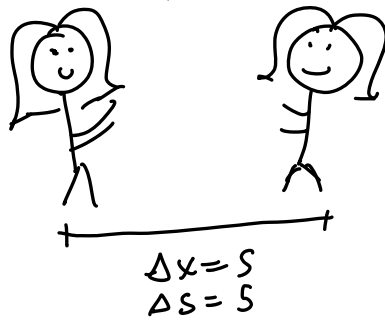
$$\sqrt{(\Delta s)^2} = \sqrt{50} \Rightarrow \Delta s = 7.07$$

$$\Delta s \approx 7$$

Without moving, the physical distance between the twins grew!



$$T = 2$$



$$T = 1$$

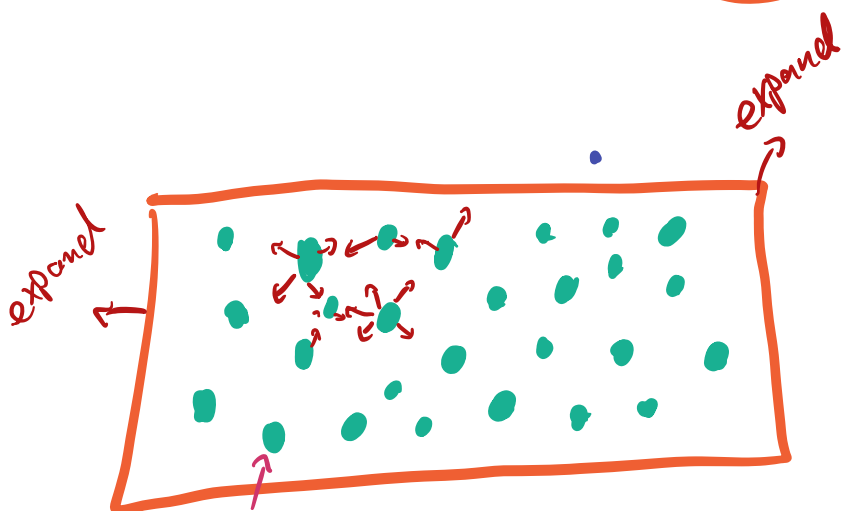
B: g
Bang →

$$\Delta x = 5$$

$$\Delta s = 0$$

$$T = 0$$

Cosmic Expansion



like a box
of gas

Instead of the
points being
atoms, they
are galaxies
all move away
from each other.

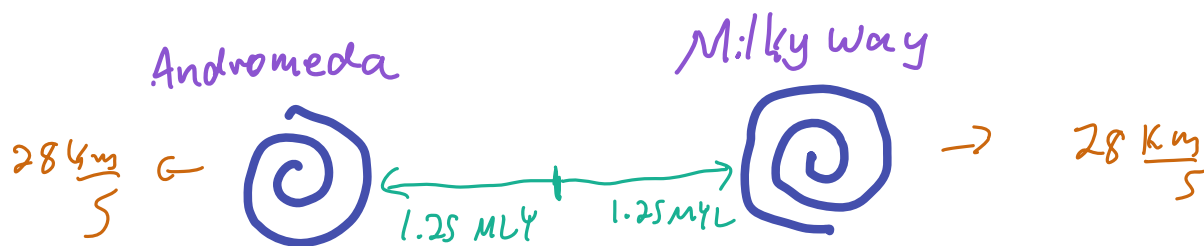
1 of about

2,000,000,000,000 galaxies
in the universe

Define: $V = H d$
 Speed of planet $\uparrow$ H $\uparrow$ distance away the planet is
 Hubble constant

$$H = \frac{\dot{a}}{a}$$

Type 1A Supernovae: $H \approx 74.03 \frac{\text{km}}{\text{s}} \text{Mpc}$



$$V = H(1.25 \text{ MLY}) = 28 \frac{\text{km}}{\text{s}}$$

History of the Universe

$t < 10^{-36}$
Seconds



Plank era
Grand Unification era

$t < 10^{-43}$
 $t < 10^{-35}$

10^{-36}
second

Inflation



E.O.M

$$\ddot{\phi} + H\dot{\phi} + V'(\phi) = 0$$

Harmonic oscillator

$$\ddot{x} + \gamma\dot{x} + kx = 0$$

Strongly damped

$$|V(\ddot{\phi})| \ll H$$

$$w = \frac{\dot{\phi}^2 - V}{\dot{\phi}^2 + V} = -1$$

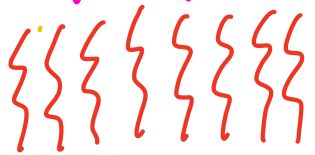
$< t <$

10^{-32}

Second

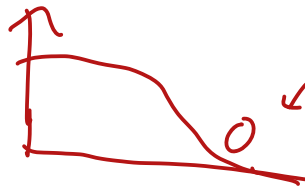
ϵ density:

$$\frac{n_p}{V} = \frac{n_{\bar{p}}}{V} \approx 0$$



$$\frac{n_p}{V} = \frac{n_{\bar{p}}}{V} \neq 0$$

reheating
 $v(\phi)$



$t < 10^{-14}$
Seconds

$$\frac{n_p}{N} > \frac{n_{\bar{p}}}{V}$$

Electroweak baryogenesis

$AL \rightarrow B$ $B-L$ symmetry

He^+ H^+

Nucleosynthesis

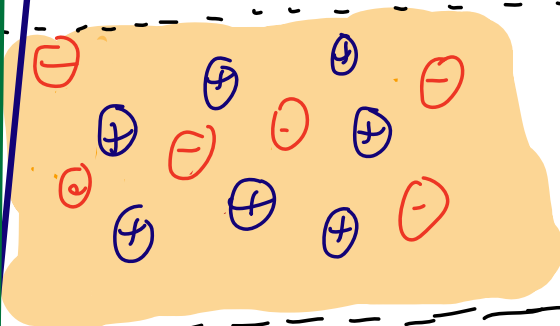
$t < 10^{-3}$
seconds

Plasma universe

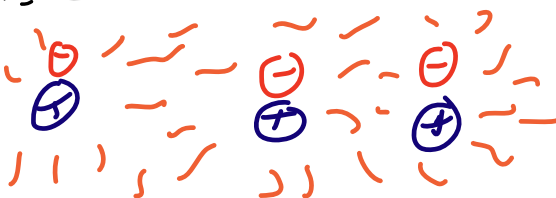
10^{-3} seconds

$< t <$

370,000
years



$t = 370,000$
years

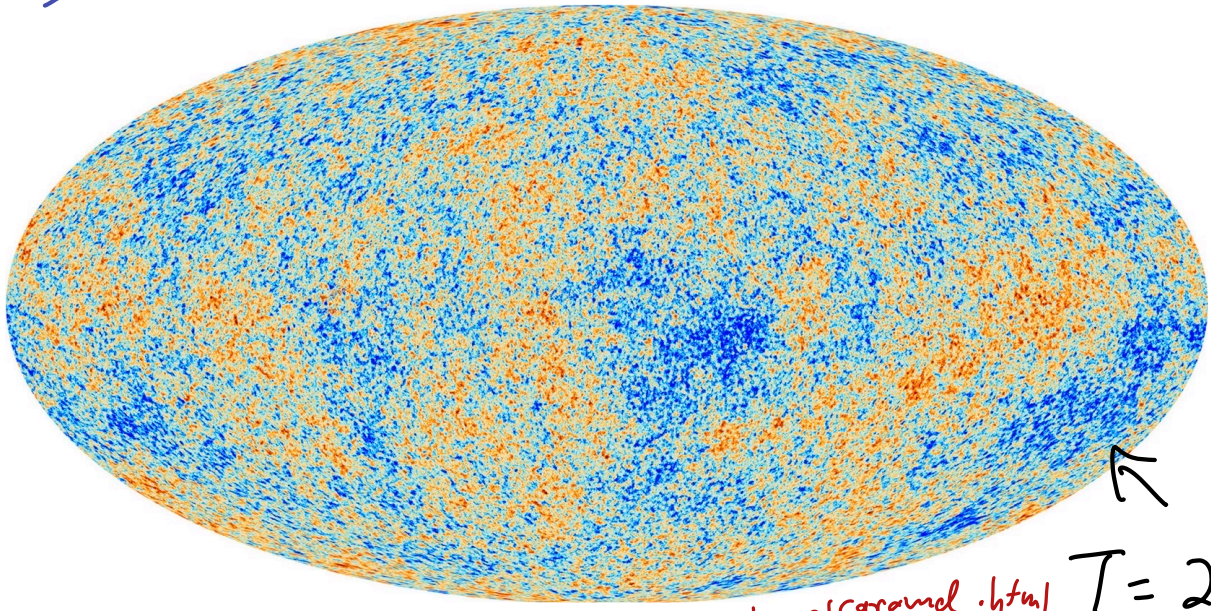


Cosmic Microwave
Background



3000 K

Cosmic Microwave Background



[Space.com/33982-cosmic-microwave-background.html](http://space.com/33982-cosmic-microwave-background.html) $T = 2.73 \text{ K}$

$$\text{Anisotropies } \frac{\Delta T}{T} = 10^{-3} \text{ K}$$

Where do these anisotropies come from?

Perturbations of Inflation

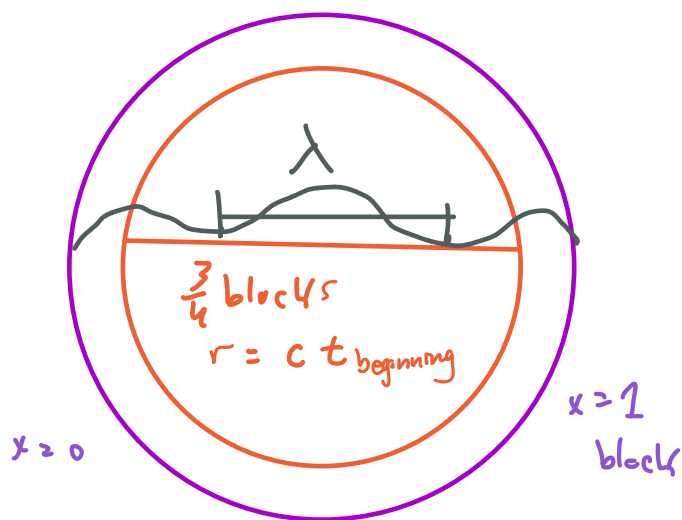
$\frac{k}{a} > H$ (wavelength is smaller than hubble radius)

$$\psi_k = c \frac{1}{a(t)} e^{i k a t} e^{i \vec{k} \cdot \vec{x}}$$

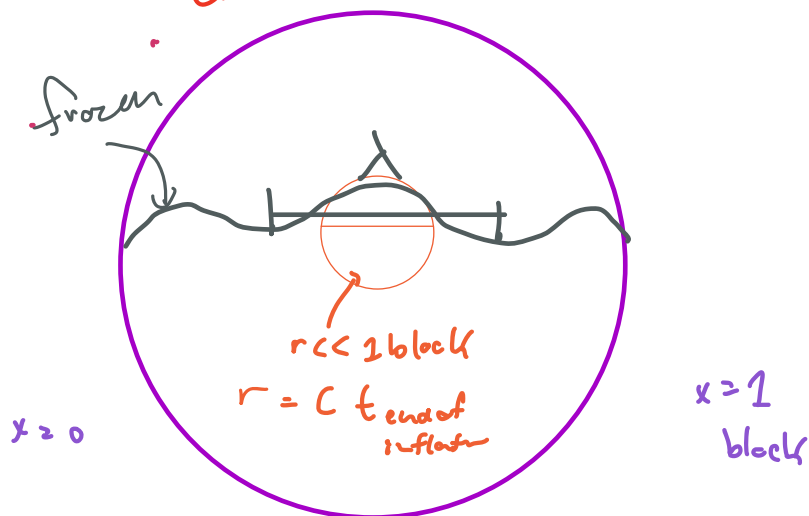
$\frac{k}{a} < H$ (wavelength is larger than hubble radius)

$$\psi_k = c \frac{H}{k} e^{i \vec{k} \cdot \vec{x}}$$

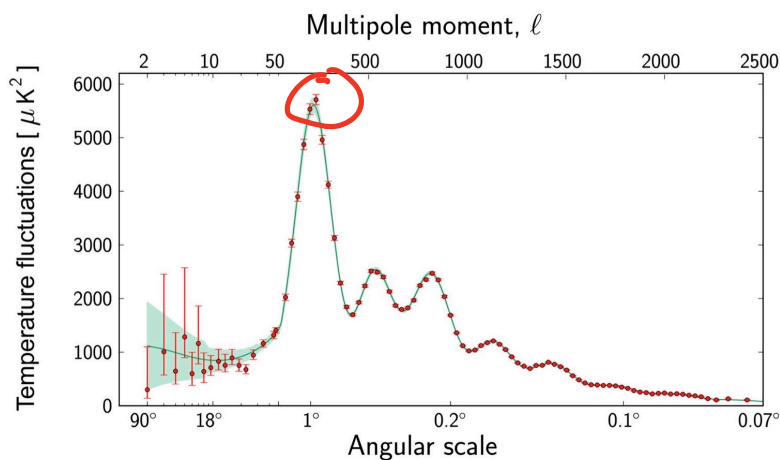
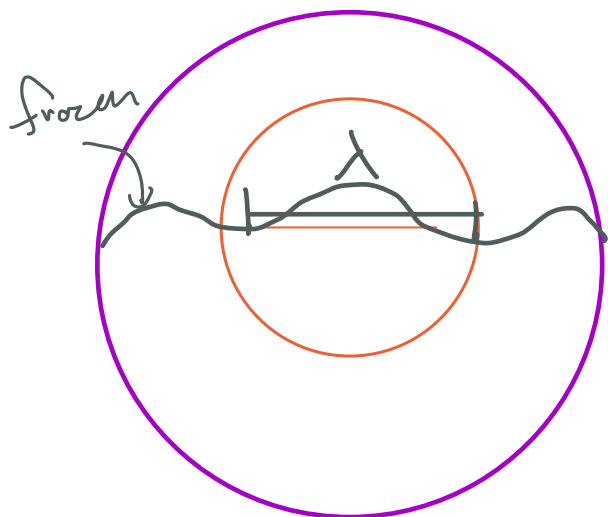
Beginning of inflation



end of inflation



$t = 370,000$ years

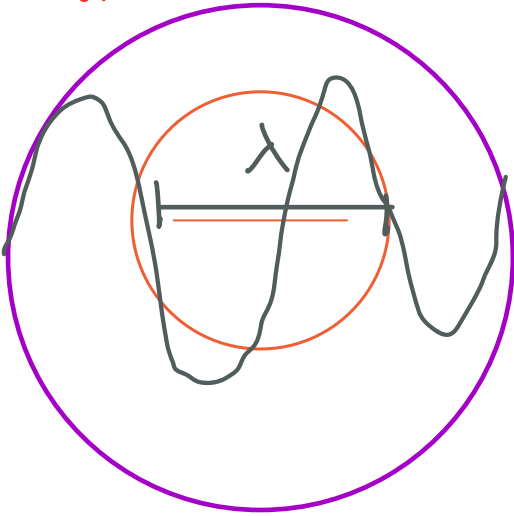


Second Peak

$\frac{k}{a} > H$ (wavelength is smaller than Hubble radius)

$$\psi_k = c \frac{1}{a(r)} e^{i k a t} e^{i \vec{k} \cdot \vec{x}}$$

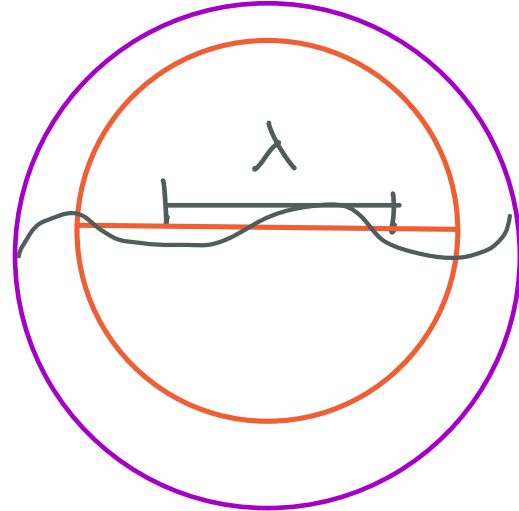
$t_{\text{end}} < t_i < t_{\text{recombination}}$



$\frac{k}{a} < H$ (wavelength is larger than Hubble radius)

$$\psi_k = c \frac{H}{k} e^{i \vec{k} \cdot \vec{x}}$$

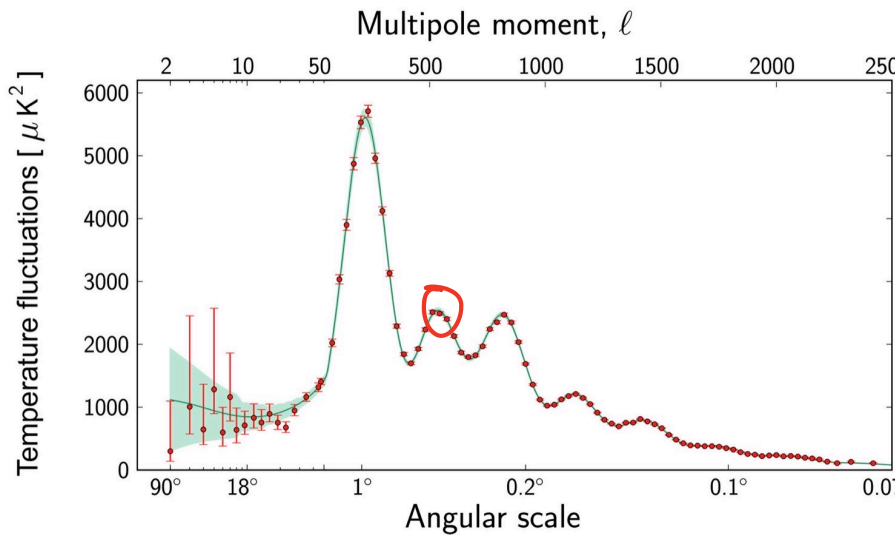
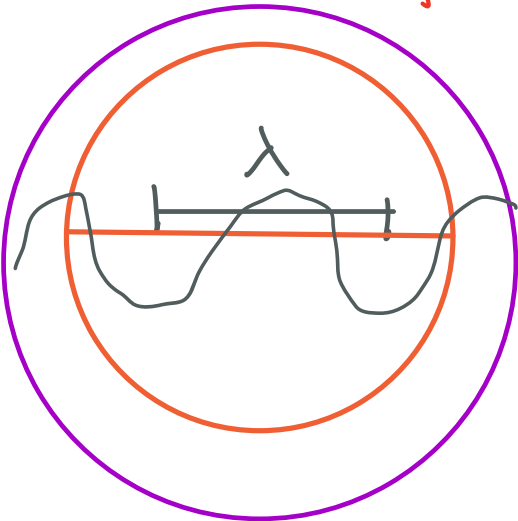
$t_i < t < t_{\text{recombination}}$

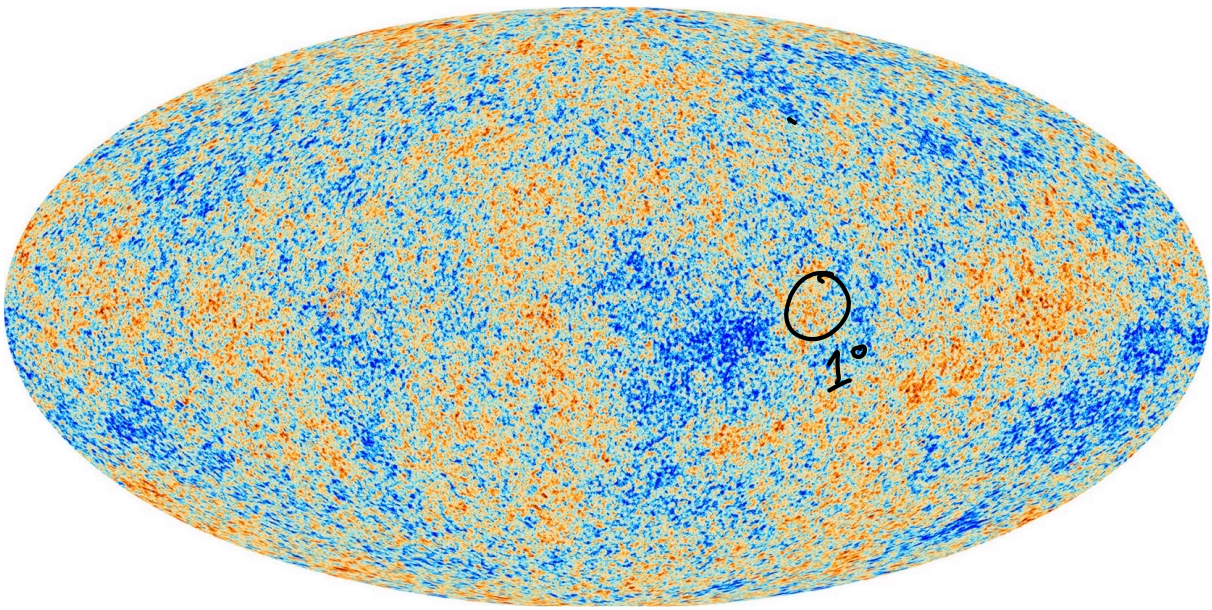


$x \approx 0$

$x \approx 1$
block

$t = 370,000$ years





Infer $H(z)$ from size of CMB Anisotropies

$$H(z) = H_0^f \left[\left(\Omega_m (1+z)^3 + (1-\Omega_m) \left[1 + \Delta \exp(- (z/z_c)^\beta) \right] \right) \right]^{1/2}$$

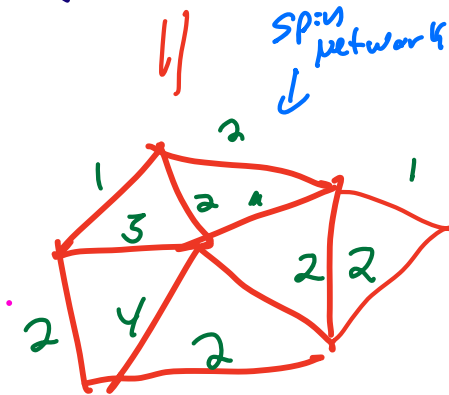
Early time : $H_0 = 67.44 \pm 0.58 \frac{\text{km}}{\text{s MPC}}$
CMB

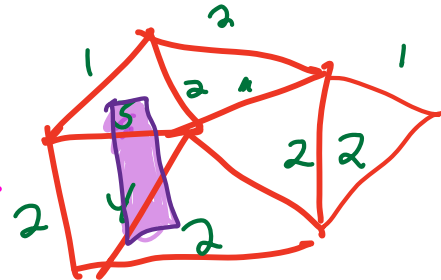
Late time : $H_0 = 74.03 \pm 1.42 \frac{\text{km}}{\text{s MPC}}$
Cepheids

Loop Quantum Gravity

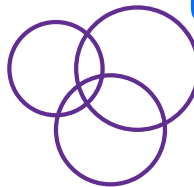
QM: $H|\psi\rangle = \left(\frac{1}{2} + n\right) \hbar \omega |\psi\rangle$

LQG: $\hat{A}|\psi_s\rangle = n \Delta l_p^2 |\psi_s\rangle$



$A|\psi_s\rangle =$  $= (3+4) l_p^2 |\psi_s\rangle$
 $= 7 \Delta l_p^2 |\psi_s\rangle$

loop Representation

 $W_r[A] = \text{Tr} \left[P \left(e^{-i \oint_{\gamma} A_a(s) ds} \right) \right]$

Wilson loop

forms a basis

$\Psi[A] = \sum_r \Psi[r] W_r[A]$

← analogous to lattice QCD

L QCD	Gravity
Physical field $e^{i\theta\gamma^a} \psi\rangle$	Metric $g^{\mu\nu} = E^\mu E^\nu$
G (non "A _n ")	ST connection $A = O(L g^{\mu\nu})$
Strong force	gravitational force

Derivative $\rightarrow$

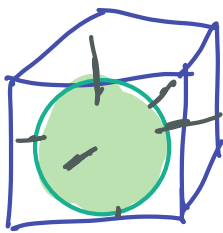
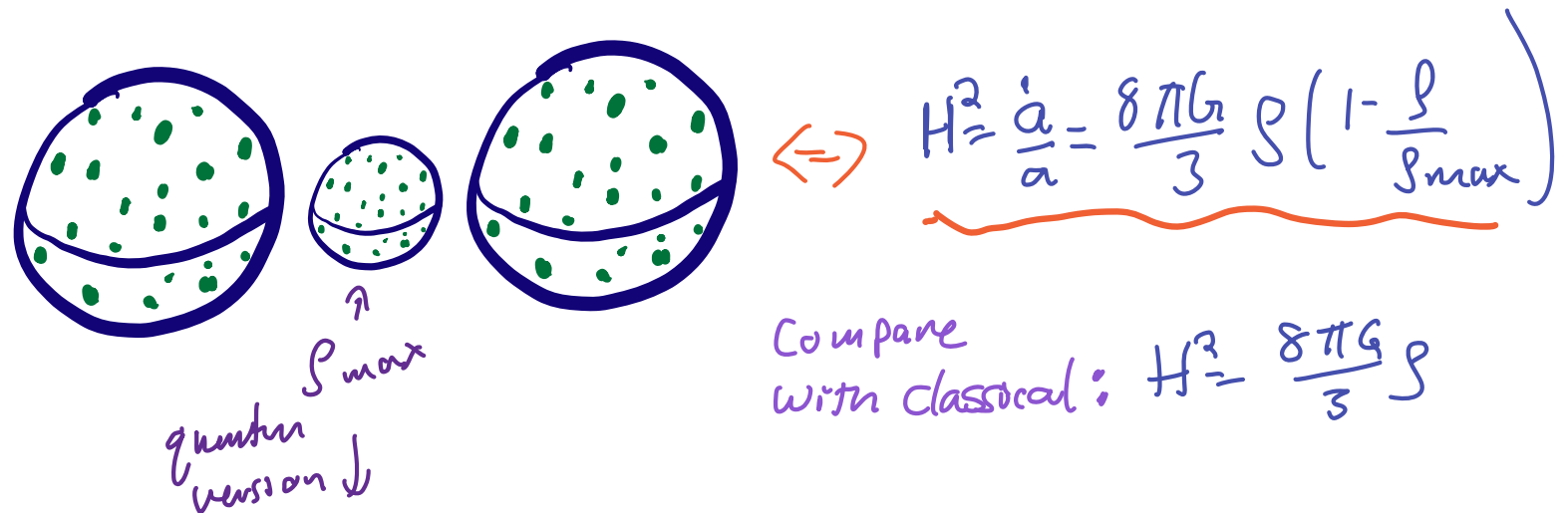
20

Loop Quantum Cosmology

Since the area eigenvalue $\neq 0$

$$\rho_{\max} = \frac{1}{4\pi\gamma^2\Delta G^2\hbar}$$

This excludes a big bang and implies a big bounce



Area of quantum bounded universe

$$n \Delta l_{Pl}^2 = 6 \Delta l_{Pl}^2$$

1.) We restrict the geometry at the bounce to be in the "ground state" to have minimum area & grav. entropy

A.L.A. Weyl tensor Vanishes at initial condition

2.) Require perturbations to have minimum uncertainty

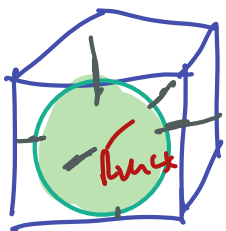
If we can ignore perturbation back reactions
 the full quantum geometry is indistinguishable to the
 effective dressed FLRW metric

$$\tilde{g}_{ab} = \sqrt{\frac{\langle \hat{H}_0^{-1/2} \hat{a}(t) \hat{H}_0^{-1/2} \rangle}{\langle \hat{H}_0 \rangle}} \left(\langle \hat{H}_0^{-1/2} \rangle \left(\langle \hat{H}_0^{-1/2} \hat{a}^4(t) \hat{H}_0^{-1/2} \rangle \right)^{1/2} dt + d\vec{x}^2 \right)$$

Quantum geometry becomes effectively encoded in
 the matter field

$$\left(\nabla^2 - \frac{\partial^2}{\partial t^2} + \frac{\hat{U}}{a} \right) \tilde{Q}_{\leq 0} \Rightarrow \left(\nabla^2 - \frac{\partial^2}{\partial t^2} + \frac{\tilde{U}}{a} \right) \tilde{Q}_{\leq 0}$$

with :
$$\tilde{U}(t) = \frac{\langle \hat{H}_0^{-1/2} \hat{a}^2(t) \hat{U}(t) \hat{a}^2(t) \hat{H}_0^{-1/2} \rangle}{\langle \hat{H}_0^{-1/2} \hat{a}^4(t) \hat{H}_0^{-1/2} \rangle}$$



New length scale :

$$k_{LQC} = \left(\frac{R_{max}}{6} \right)^{1/2} = \left(\frac{62}{6 l_{Pl}^2} \right)^{1/2}$$

Perturbations with $k < k_{LQC}$
 become excited

Modified Mukhanov - Sasaki Mode:

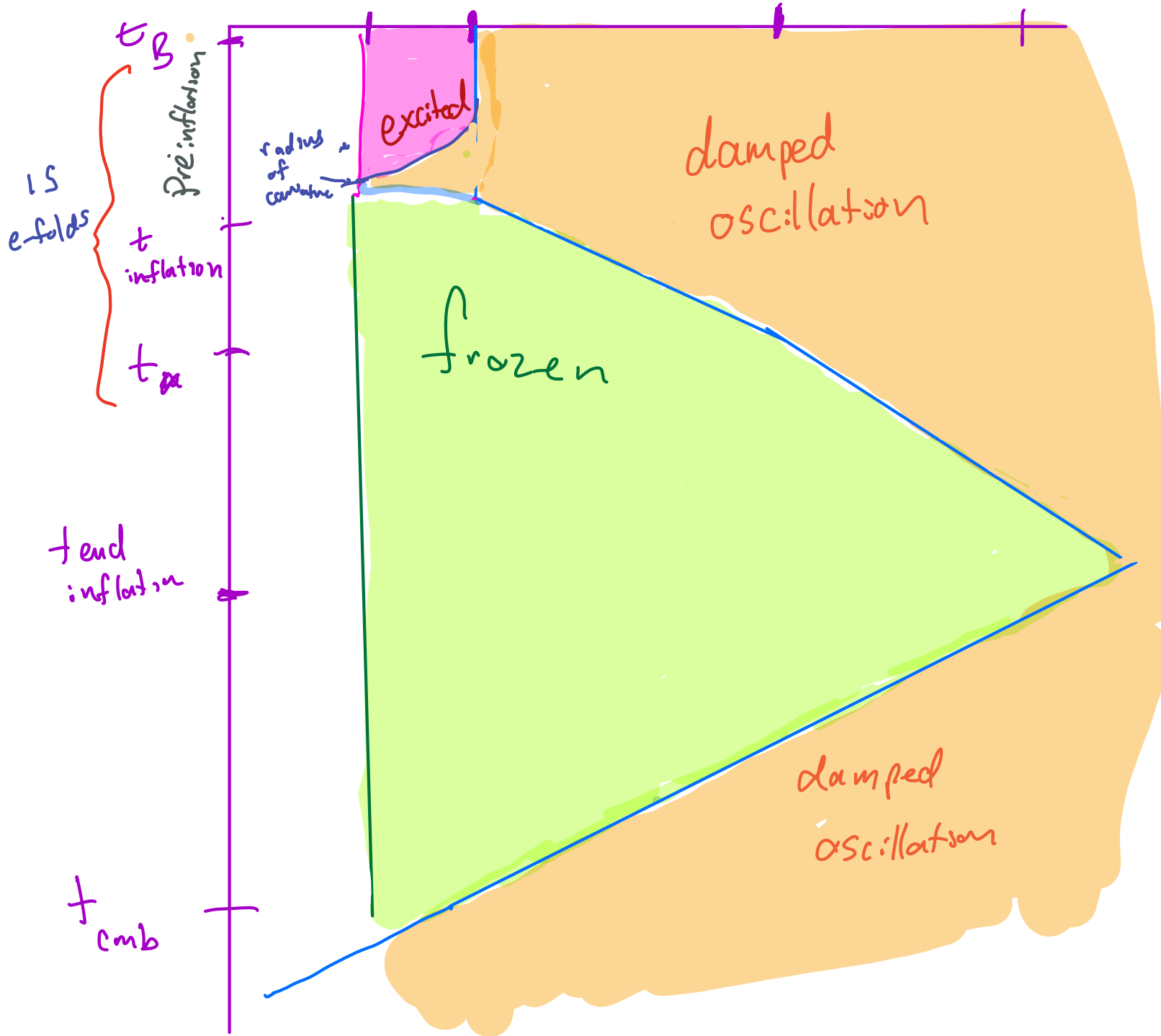
$$\partial_{\eta}^2 \chi_k = a^2(\eta) (1^{-2} - (6\epsilon_{\text{end}})^{-2}) \chi_k$$

hubble length line

$1/k_{*}$ k_{LQC}

k_{*}

$300k_{*}$



Observational Consequences of LQC pre-inflationary perturbations

The standard Ansatz PS: $P_R(k) = A_s \left(\frac{k}{k_*} \right)^{n_s - 1}$

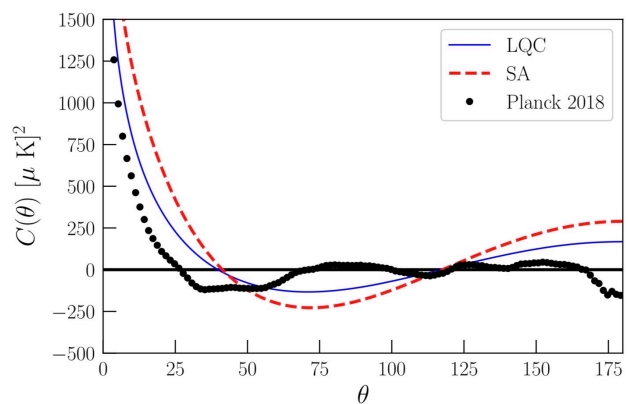
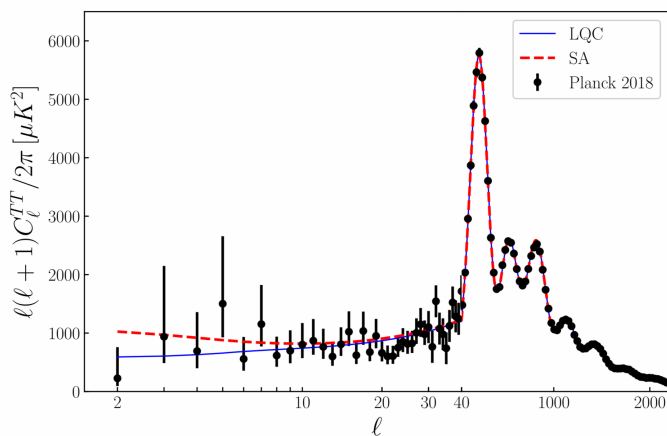
LQC Modification: $P_R^{LQC}(k) = f(k) A_s \left(\frac{k}{k_*} \right)^{n_s - 1}$

$f(k) \approx 1$ for $k \gg k_{LQC}$

but $f(k) < 1$ for $k < k_{LQC}$

C^{TT} power suppression

large scale power anomaly



Ashdelfer 2020

Questions ??